Fibre Assurance & Network Operations

Product Overview

Telecom NOCs monitor optical equipment, GIS maps, OTDR testers, and change tickets in separate tools. We bring them into one view so operators can spot fibre threats early, send repair crews to the exact street location, and verify backup routes in seconds.

The Problem Today vs. What Changes

WHAT HAPPENS TODAY	WITH OUR PLATFORM
No warning before a cut. Teams only react after an excavator snaps a buried cable.	**Early digging alert.** Spots ground vibration on live fibre and shifts traffic before the break.
Crews dig 200m off target. Raw OTDR distance counts spare fibre coiled inside manholes.	**Exact street GPS.** Subtracts manhole slack loops on the GIS map to pinpoint the trench spot.
Wasted field dispatches. Customer power blackouts and rain fade look like fibre cuts.	**False-alarm filtering.** Cross-checks power signals, weather, and active maintenance tickets first.
Hidden shared risk. "Backup" enterprise links quietly share the same duct or bridge.	**Instant route check.** Confirms zero shared ducts or chambers before quoting a customer.

How It Works End to End

STEP 1

Ingest

Reads live alarms, OTDR traces, GIS ducts, and change tickets (read-only).

STEP 2

Isolate

Groups thousands of alarms into one root cause and rules out power or weather.

STEP 3

Act

Suggests a clean backup path and exact GPS dig spot for engineer sign-off.

STEP 4

Verify

Checks splice quality remotely before closing the ticket and drafting the RFO.

Six Capabilities We Deliver

1. Pre-Cut Digging Warning	Detects heavy digging near buried ducts and moves traffic to a backup path before the fibre breaks.
2. Pre-Sales Diverse Routes	Checks two customer sites on a map in seconds to guarantee zero shared ducts, manholes, or bridges.
3. Subsea Vessel Watch	Tracks ships over coastal cables to flag anchor drag early and separate subsea faults from land cuts.
4. Maintenance Match	Flags immediately when an outage overlaps an active change window or a nearby municipal roadwork permit.
5. Repair Time & SLA Shield	Estimates splice time by cable size, writes EN/Arabic incident reports, and flags outages exempt from SLA credits.
6. Field View & Splice Sign-Off	Gives crews exact fibre color codes on mobile and runs a remote OTDR test to confirm a clean repair.

Where It Fits Best

Metro & Dark Fibre (e.g., Constl): Fewer urban excavation outages and instant diverse-route quotes for banks and hyperscalers.

National Telcos (e.g., e&, du, Vi): Cuts alarm noise across optical and 5G, stops wasted truck rolls, and automates bilingual SLA reports.

Subsea & Wholesale (e.g., PCCW Global): Early anchor-drag visibility at cable landings and faster wholesale incident handoffs.

Low-risk rollout: Connects read-only to existing optical NMS (Ciena, Nokia), OTDR testers (VIAVI, EXFO), GIS maps, and ServiceNow/Jira. A typical pilot runs for **4 weeks on a single metro ring** with full human NOC approval on every action.